

Mission 3 – Wind Simulation

Objective

To evaluate drone stability and path tracking under simulated wind conditions using the Mission 1 waypoint setup, with specific analysis of how wind direction and speed affect resultant velocity, ground track, and attitude corrections.

Parameters Used

- SIM_WIND_SPD = 15 mph / 45 mph
- SIM_WIND_DIR = 315° (Northwest)
- SIM_WIND_TURB = 0
- WPNAV_SPEED = 30 mph (airspeed command)
- Default PID tuning parameters

Resultant Velocity Analysis

Wind from the Northwest (315°) pushes the drone toward the Southeast. This creates three distinct aerodynamic conditions depending on the drone's heading at each leg of the mission. On **headwind legs** (drone flying Northwest, into the wind), the wind directly opposes forward motion, reducing ground speed below the commanded airspeed. On **tailwind legs** (drone flying Southeast, with the wind), wind adds to forward motion, increasing ground speed above the commanded value. On **crosswind legs** (drone flying Northeast or Southwest), the wind pushes the drone laterally off the intended track, causing **crabbing** — where the drone's nose points in a different direction than its actual ground track in order to maintain the correct path. The autopilot compensates by angling into the wind, so the resultant ground velocity is the vector sum of the commanded airspeed and the wind velocity acting on the airframe.

Wind: 15 mph (Northwest)

Observations:

- Drone was slower when flying against the wind (headwind legs)

- Slight shaking and oscillations visible throughout
- Minor roll and pitch corrections observed
- Path was still followed successfully

Concrete Example from Flight: During the leg heading into the Northwest wind, ground speed dropped to approximately **29 mph** despite a commanded airspeed of 30 mph — representing nearly a 50% reduction in over-ground progress due to the headwind. On the return tailwind leg, ground speed recovered noticeably above the commanded value.

Explanation: At 15 mph wind, the wind force was weaker than the drone's navigation speed, so the autopilot could compensate by tilting into the wind. Crosswind legs produced visible crabbing — the drone's nose was offset from the direction of travel — but the flight controller successfully maintained the ground track through continuous attitude corrections. Minor oscillations reflected the PID controller actively fighting the wind load.

Wind: 45 mph (Northwest)

Observations:

- Drone exhibited severe shaking throughout the mission
- Failed to follow the programmed path accurately
- Significant drift in the Southeast direction (downwind)
- Large, rapid attitude corrections visible in roll and pitch

Concrete Example from Flight: On the headwind leg into the Northwest, ground speed fell to approximately **0.8 mph** — the wind nearly cancelled all forward progress. On crosswind legs, the drone was visibly pushed off the planned track with lateral drift exceeding the corridor, and crabbing angles became extreme as the autopilot attempted to compensate.

Explanation: At 45 mph, wind speed exceeded the drone's commanded navigation speed of 30 mph. On headwind legs, the drone could not make meaningful forward progress. On crosswind and tailwind legs, the wind dominated the resultant velocity vector, pushing the drone off the planned ground track faster than the flight controller could correct. The propulsion system reached its compensation limits, causing unstable flight and mission path deviation. This represents a condition beyond the drone's operational wind envelope.

Conclusion

- At 15 mph, the drone maintained control with minor instability. Headwind legs showed reduced ground speed (~29 mph vs. commanded 30 mph), crosswind legs produced measurable crabbing, and tailwind legs showed speed increase — all consistent with expected resultant velocity behavior.
- At 45 mph, wind force dominated the resultant velocity on all legs. Ground speed on headwind legs dropped near zero (0.8 mph), crabbing angles became uncontrollable, and path deviation confirmed the drone exceeded its stable operating wind limit.
- This demonstrates that wind not only affects speed but fundamentally alters the drone's ground track through vector combination of airspeed and wind velocity — and that beyond a threshold wind speed, the flight controller cannot maintain mission accuracy regardless of PID corrections.